

Living to Ninety

7

LIVING TO NINETY

It is a popular belief, but a fallacy, that we can learn about what contributes to extreme old age by studying very old survivors. This certainly needs to be done, but it is also necessary to know about the characteristics of other members of their birth cohort, in early childhood, adulthood and early old age.

—M. BURY AND A. HOLME, *Life after Ninety*

IN MY PREOCCUPATION with love, joy, and relationship in old age, I tend to forget the more mundane but absolutely crucial importance of staying alive. Physical health is just as important to successful aging as social and emotional health. In 2011, the Grant Study and the Lothian Birth Cohort 1921 became, as far as I know, the world's first prospective studies of the physical health of nonagenarians. ¹ (The Lothian Birth Cohort was formed in Edinburgh in 1932. It included more than 80,000 eleven-year-olds, many of whom have been followed over the years to the age of ninety.)

In this chapter I will focus on the sixty-eight surviving members of the Grant Study who are still active as of March 2012, but I will also review what happened to the men who died along the way. This exploration will take me through several important revisions of the received wisdom of earlier years, including some once convincingly backed up by hard data. There were surprises involved, and some shocks, too, for argument and theory can never be enough. We need documentation and proof. In the study of lifetimes this means prolonged follow-up and systematic retesting of serious hypotheses—not over a year or two, but over decades. That's the very first and most fundamental lesson of the Harvard Study of Adult Development, and the one on which all the other lessons depend: While life continues, so does development. The study of very old age is the ultimate case in point.

In 1980, Stanford University internist and epidemiologist James Fries recognized that modern medicine was not extending the human lifespan, and yet survival curves were changing. More people were living vitally until eighty-five or ninety, and then dying quickly, like the wonderful one-hoss shay in Oliver Wendell Holmes's poem, which ran perfectly for a hundred years and then fell apart all at once. ²

Fries called this phenomenon "compression of morbidity." ³ In 1900, because most deaths were premature, the human survival "curve" was a diagonal line; now it is more of a rectangle—especially if you have no risk factors (Figure 7.1). In 2040 there will be ten times as many eighty-five-year-olds as there were in 1990. This is not because the normal human lifespan is any longer than it was, but because fewer people will die before eighty. After eighty the lifespan will reflect little increase. Medical advances like antibiotics, new cancer treatments, and kidney transplants all serve to decrease premature death. But they do not alter the fact that the bodies of most of us, like the one-hoss shay, have not evolved to live past one hundred.

In 2012 the longevity of the Grant Study men, who were selected for physical health, is clearly ahead of the historical curve. While only about 3 percent of the white American males born in 1920 are expected to celebrate ninetieth birthdays, 77 (28 percent) of the 268 Study men already have, and 7 (3 percent) more are eighty-nine and likely to join them. Their projected age of death is already four years longer than that for white males born in 2009. ⁴ (By way of comparison, only about 18 percent of the gifted Terman men made it to ninety. ⁵) One purpose of this chapter, therefore, is to give readers a view into the future, to a time when living to ninety will have become more commonplace.

Figure 7.1 Percentage of men alive at each age.

Some may ask: How long do we want to live? Our society stereotypes the very old as frail, ill, and unhappy. No one likes the prospect of long years confined to a nursing home, helpless, mindless, or in pain. And it can't be denied that aging means loss. Our bodies begin their long, slow process of failure at thirty; by age seventy we can identify only fifty percent of the smells we recognized at forty. Our vision declines until by eighty few of us can safely drive at night. ⁶

Even among the surviving Grant Study nonagenarians, 25 percent are cognitively impaired.

But that means that three out of four Grant Study ninety-year-olds are still cognitively intact. Although we often hear that our brains begin to shrink at twenty and that we will have lost fully 10 percent of our brain cells by seventy, modern brain imaging techniques suggest a less grim picture. Normal brain shrinkage is less than we feared, and estimates so far may also reflect the practice of averaging individuals with specific brain-destructive pathologies—Alzheimer’s, trauma, alcoholism—in with everyone else. 7 In addition, we can speculate that some lost brain matter is the result of judicious “pruning” of cells that are no longer in active use; after all, due to such pruning, our brains contain fewer than half the synapses at twenty-one than they do at five. 8 There may not be as much cause for despair as we fear.

Furthermore, the mental life of the elderly is not as bleak as we are sometimes given to believe. After forty, the fear of death declines steadily and belief in an afterlife becomes more common. Careful epidemiological studies from multiple centers reveal that depression among the elderly does not increase. 9 Indeed, recent studies show that older adults report less depression, lower levels of negative affect, and more assertiveness and positive emotion. 10 As one Terman great-grandmother put it, “I hate my waist, but I love my psyche!” Most of the Grant Study men who reached their eighty-fifth birthdays were glad they did.

The most authoritative study of old age available is the MacArthur Foundation study, which is summarized in Rowe and Kahn’s *Successful Aging*. 11 That study refutes the specter of dismal years in a nursing home, finding that the average eighty-five-year-old man will spend only about six months in an institution before he dies, and the average eighty-five-year-old woman about a year and a half. People who live to be one hundred, those data show, are usually pretty active at ninety-five.

In the Grant Study men, in fact, for a long time there seemed to be so little cognitive decline that we did not even begin testing them for cognitive function until they were eighty. At that time, 91 percent of them tested at normal levels. Of the seventy men who have so far survived to reach ninety, fifty-eight were cognitively intact at that age, and had showed over the preceding ten years an average decline of only one point on the 41-point Telephone Interview for Cognitive Status (or TICS). 12 (I’ll say more about the TICS shortly.) Almost three-quarters of the Grant Study ninety-year-olds were as sharp as ever, if just a bit slower. It’s true that a majority could no longer give the full name of the vice president: “I know he was a senator from Delaware. . . .” Yes, finding names can be a problem after seventy. So can finding your car in a parking lot!

The MacArthur Study supports Fries’s view that the years of disability are growing shorter even as we live longer. Bury and Holmes report that chronic pain declines from a high of 28 percent at eighty-five 13 And while dementia is to only 19 percent after ninety-five.ghastly for afflicted individuals and their families, in the Grant Study, only ten of the forty-one men who became demented before eighty-nine lived for longer than five years after their dementia was recognized. Moreover, some, although not all, epidemiologists believe that the incidence of Alzheimer’s declines after ninety-five. 14

Progressive diminution of physical reserves is an inevitable part of aging, but the rate at which this happens varies. Forty percent of the general population remains “fully functional” after eighty-five. That is a hard criterion to define, but as Table 7.1 illustrates, about 55 percent of the Grant Study men between eighty-five and ninety appeared to meet it. And only 9 percent of them were confined to home or wheelchair.

Table 7.1 Instrumental Activities of Daily Living at Age 88 ± 1 (N=60 *)

* Sixty people returned this questionnaire.

The questionnaires that the most energetic of the men returned at ninety (or almost) convey some of the flavor of their lives. One man who had been divorced four times described the happiness of his fifth marriage—which had lasted for twenty years—as 7 out of 7. He reported no cutbacks in any of his physical activities, except that he had given up night driving. For his energy level he checked “vigorous”; he continues to play tennis. Another man was still earning \$60,000 a year at eighty-seven, and also engaging in thirty hours a week of volunteer activity. He, too, described his marriage as 7 out of 7, and at ninety he was sexually active. A third man exercised fifteen or more hours a week, was sexually active, and gave his marriage 6 out of 7 points. He engaged in four volunteer jobs.

TWO LIVES

The life of Alfred Paine. The realities of the men's health and their subjective perceptions of it did not always coincide. Good self-care, high morale, intimate friends, and robust mental health often make the difference between having an illness and feeling sick. Let me now therefore give a more nuanced portrait of both ends of the aging spectrum, starting with Alfred Paine (previously discussed in *Aging Well*), who was both ill and sick, as per my differentiation above. His greatest strength was that he did not complain; his greatest weakness was that he knew himself very little. He could not acknowledge either his alcoholism or his depression. Paine had one of the best scores in the study on a subjective inventory of depressive symptoms. He had never sought psychotherapy; none of his doctors had ever called him mentally ill. On questionnaires, he described himself as close to his children and in quite good physical health. It was only by interviewing him personally, talking with his wife, examining his medical record, recognizing the disappointment in his children's questionnaires, and—finally—reading his obituary that Alfred Paine's misery could be fully appreciated.

His ancestors had been successful New England clipper-ship captains. One grandfather was a merchant banker; the other was a president of the New York Stock Exchange. His father had graduated from Harvard and his mother from a fashionable boarding school. Paine himself arrived at Harvard with a handsome trust fund. But his childhood received the lowest rating for warmth of any man's in the Study, and in the Grant Study, at least, parental social class does not guarantee successful aging. Paine's story illustrates that money doesn't, either.

At college, Paine was often in love. But to the Study staff it appeared that to him, "being in love" meant having someone to care for him. His multiple marriages were all unhappy, especially according to his wives. This was partly because of his alcoholism (which he denied), and partly because he was afraid of intimacy.

Paine described his physical health as "excellent" at sixty-eight; but objectively it wasn't. He was seriously overweight, and he had obstructive pulmonary disease as a result of lifelong smoking. By age seventy, gallstones and an ileostomy for diverticulitis of the colon were adding to his medical woes. When I interviewed him at seventy-three, he seemed ten years older than his age—like an old man in a nursing home. He had lost all his teeth; his kidneys and his liver were failing; he had a mild dementia as the result of a drunken automobile accident. He took two medicines for high blood pressure, two for diabetes, one for gout, and one for depression. There was no question that he was physically disabled.

Paine's Decathlon score was 0. Alone of the men who returned the Study questionnaire at seventy-five, he refused to fill out the Life Satisfaction Chart that we included with it. But his joylessness was evident. I always found it exhilarating to listen to men who were zestfully mastering life's slings and arrows, but interviewing Paine made me feel very sad. In his questionnaires, he said nice things about his children. But when I asked him what he had learned from them, he responded irritably, "Nothing. I hardly see my children." One daughter said in her questionnaire that she saw her father only every three years, while the other saw him once a year and viewed him as having lived "an emotionally starved life." Alfred Paine's only son said that he had never been close to his father.

Paine's third wife was protective and loving toward him, but he was disrespectful to her, and not very caring. I asked him once how he and his wife collaborated, and he replied, "We don't. We lead parallel lives." It doesn't help to be loved if you can't let yourself feel it.

When I asked Paine at seventy-three to describe his oldest friend, he growled, "I don't have any." He could climb stairs only with difficulty and he had great difficulty walking even 100 yards; he couldn't drive at night, and he had had to give up golf because of his gout. Both his wife and his doctor saw him as seriously impaired. Yet on his last questionnaire, in a shaky hand, the seventy-five-year-old Paine called his general health "very good" and reported that he had no difficulty with physical activities. The very next year he was in a nursing home; a year after that his multiple illnesses had killed him at an age when two-thirds of his Study-mates were still alive. Paine's way of dealing with trouble was to wear blinders all the time—a striking contrast with the always-game Boatwright and Adams, who relied on rose-colored spectacles, but mostly to help themselves keep up the good fight.

The life of Daniel Garrick. I never met Daniel Garrick, although I tested his cognitive function by telephone three times—when he was eighty-one, eighty-six, and ninety. I was impressed that he did better at eighty-six than at eighty-one. When I called that middle time, Garrick told me that I was lucky to have gotten him on his day off; he worked

the other six. He was very excited about having just taught his first course at the San Francisco Art Museum—on American painters, complete with his own slides. I chose him as a counterpoint to Alfred Paine’s early decline and miserable last years on the basis of those phone calls, and because I knew that at ninety-five he was the oldest living man in the Study. However, I didn’t know much about his life before he turned eighty, so I had to read his record from the beginning.

Garrick was the oldest of six children. He grew up during the Depression in a family whose yearly income per person was \$400—that was \$3,200 per year total for a family of eight. His father, an accountant with a high school education, was often unemployed, and couldn’t understand his son’s decision to go first to acting school and then, at twenty-five, to Harvard. Garrick’s father was a very conservative man who read only the Reader’s Digest, but his mother was a sensitive, artistically inclined liberal, and Daniel modeled himself on her. The Study raters considered his childhood average for warmth.

Daniel was bottle-fed. He was fully toilet-trained by one. From the beginning, he was a “good-natured” and “very friendly” child. In school he got bullied by older children, was completely incompetent at all sports, and failed algebra. But in the ninth grade he had his first success in a school play. He loved the applause, and from then on he devoted himself to school theatricals, at which he excelled. Upon graduation from high school, he enrolled in drama school. He rejected New York’s bohemian life and at that time didn’t drink or smoke, but he took part in summer stock productions every July. He only wanted to act in plays by serious playwrights like Shakespeare and Ibsen, and he had an idealistic wish to invite the public (perhaps primarily his philistine father) into the aesthetic realm that he and his mother inhabited with such pleasure. He wasn’t able to support himself as an actor in Depression-era New York City, however, and he came reluctantly to the conclusion that he was too emotionally inhibited to be a successful actor, despite his passion for the theater.

Garrick resigned himself to a future of stage managing and teaching drama in some small college. To this end he put himself through Harvard without family help or scholarship support. A private mental hospital gave him room and board in return for working the graveyard shift as a psychiatric attendant. He would sleep from 6:00 p.m. to midnight, cover his shift (which gave him some time to do his homework), and then bicycle the five miles to Cambridge at 8 a.m. to save the carfare. Reading this tale of an iron constitution, I found myself wondering about the truism that exercise leads to long life. Maybe it’s the other way around. Maybe some people are just born with the stamina to live to ninety-five, and that stamina makes them good exercisers, too.

Even with his great physical toughness, Garrick’s college life was difficult. He had only five dollars a year to spend on clothes. There was no time for a social life—no dances, no dates, no sports. He was chronically tired and unable to earn better than C’s until he married, just before his senior year. Once his wife, a young summer stock actress, started helping with expenses, he received honor grades.

The Study staff did not all relate to Garrick. He received one of the top ratings in the Study for being “well-integrated” and “self-starting,” and one observer appreciatively called him an “intense fellow.” But when he was twenty-six, the Study psychiatrist summed him up this way: “He was chosen for the Study for outward manifestations of good health and stability. He has actually attained only very modest success in the field he has sought. . . . Although he has some insight and is making progress in understanding himself, he does not yet, and probably never will, reach the understanding necessary to be a happy person.” The Grant Study was not always good at understanding artists—it rejected Norman Mailer and Leonard Bernstein!—and despite his manifold successes, Garrick’s score on the Decathlon was only 3—average.

A decade after he joined the Study, Garrick was given a “D” for personality stability, a “D” for mental and vascular health, and a “B-” for projected longevity. Even Clark Heath called his health only “fair.” The case discussant dismissed Garrick’s wish to share his cultural pleasures as the “prestige motivation” of a show-off. Another staff member predicted “character disorder.” The Study staff of the time expected people to aim at business, law, or medicine, and there definitely seems to have been a blind spot for artistic types.

Garrick was thirty-three when his local repertory theater went bankrupt. “My nerves were torn to shreds,” he wrote. Still too emotionally constrained to act, he gave up theater work as too stressful, and returned to a safer goal. At forty he saw himself as “mediocre and without imagination,” but he acquired the necessary Ph.D. and set out to make his way

teaching drama and theater history in the “small college” of his imagination.

When I got to this point in my reading, I began to fear that my memory was playing tricks on me. Was I erroneously recalling the delightful telephone conversation of ten years before? Was my exemplar nonagenarian really a depressed narcissist? I didn’t yet know that his doctoral dissertation had been published and won prizes, or that, when he was eighty-eight, a grateful former student gave his college a substantial gift in Professor Daniel Garrick’s name. All I could see in Garrick’s future was years of disappointment. Like Garrick himself, I viewed him as a failure.

My anxiety drove me to cheat. I opened his latest Study folder, where the first thing I saw was a newspaper clipping. Actually it was more than a clipping; it was the whole front page of the San Francisco Chronicle’s Sunday Arts section. The headline read: “For Daniel Garrick, 89, It’s Always a Full House.” I heaved a sigh of relief and went back to the chronological story. As Freud once said, “Before the problems of the creative artist, analysis must, alas, lay down its arms.”¹⁵ Clearly I had more in common with those early Study investigators than I wanted to acknowledge. I needed a reminder that artists need time to mature. Another reason for longitudinal studies. . . .

Garrick went through a lot of occupational and marital unhappiness in midlife, and both the Study staff and I dismissed him as a failure on the strength of it. He had given up his career of choice and settled for second best. At fifty-three he separated from his first wife, and his sex life for a while was so uninspired that he turned to archery in sublimation. But as it turned out, those setbacks were temporary. Garrick spent the second half of a very long working life in one distinguished repertory company after another. At eighty, he had long stopped needing archery; many fifty-year-olds would have envied him his passionate marriage. Daniel Garrick was not the only Grant Study man who rediscovered in middle age a life of feeling that had been left behind in childhood. If women, at least until recently, have been socialized into giving up vaulting ambition at menarche, boys in grammar school are encouraged to abandon right-brain heart for left-brain reason. If they’re lucky, the tide may shift in maturity, so that they can develop conscious emotional lives. I can never resist quoting E. M. Forster in this context: “Only connect the prose and the passion and both will be exalted.”¹⁶

As he got older, Garrick finally outgrew the emotional inhibitions that had so plagued him as a young actor. Neuroscientist Francine Benes has demonstrated that with maturity the frontal cortex of the brain, the executive center, becomes more securely attached to the brain’s emotional center, the limbic system.¹⁷ In his early fifties, Garrick quit his “day job,” giving up his tenured drama professorship to become a paid actor in a Shakespearean repertory theater out West. That is the point at which I’d say he achieved true Career Consolidation, although even before that he had already received standing ovations in an off-Broadway *King Lear*. About that performance, one reviewer wrote: “Garrick is the greatest Lear I can imagine. He appears to be the living man, in all his arrogance, petulance, fear, and confusion. A truly unforgettable experience.”

Daniel Garrick had matured emotionally to the point where he could now support himself by sharing not just his ideas, but his feelings, with an audience. He no longer needed to separate them, as young people, defensively, often do (see Chapter 8). He had also learned how to play; I think of Marlon Brando as the semi-retired Godfather, chasing his grandson joyfully around his garden. At fifty-five Garrick took his first vacation ever. After the long sedentary years of teaching, he began jogging and returned to long bicycle rides. And there’s the chicken-egg question again. Was he unusually healthy because of all the bicycle touring at fifty-five? Or was the bicycle touring possible thanks to an unusual endowment of good health?

At seventy-six, Garrick had a stint of alternating performances of *King Lear* with a more famous actor. A reviewer wrote, “I found Garrick’s interpretation the stronger in many ways. . . . his fury, his self-pity, his pain, and his madness more vital and deeper reaching. I feel very fortunate to have seen him.” Garrick would eventually play opposite Gwyneth Paltrow, Lynn Redgrave, and Olympia Dukakis.

Garrick got married again at seventy-eight, to his longtime companion. When he was eighty-six, Maren Batalden wrote, “Garrick. . . is thoughtful, intelligent, warm, self-aware, humble, curious, engaged, emotionally present, enthusiastic, honest. . . . He is extraordinarily successful in both love and work. His marriage to Rachel is clearly a deep, joyful connection that sustains him, surprises him, and makes him weep with gratitude. The marriage is remarkable for the freedom each has given the other to be independent—they each have separate spheres, their own friends and interests and social styles, but nevertheless enjoy each other’s company and love each other’s bodies.” It’s wonderful to hear that

bodily love can arise de novo so late in life, and exist so passionately after eighty. Discoveries like that require longitudinal study.

Batalden continued, “In many ways, these seem to be the best of times for Daniel Garrick—he now relishes the fruits of his discipline. He loves loosely and well, works passionately, keeps his mind and heart open and engaged, makes new friends, relishes his hobbies, takes real nourishment from the past, permits himself a wide range of deep sorrow and great joy, adapts stoically to the inconveniences of aging in his body by staying focused on the things that matter to him.”

Since age eighty, Garrick has enjoyed the highest possible mood scores. He did give up biking at seventy-six due to multiple aches and pains; nevertheless, at eighty-six his only medicine was Viagra “as needed”—twice a week, he boasted. Yes, he had begun to feel his age, but for the next five years he kept up with the demands of a major Shakespearean theater, memorizing his lines and rehearsing, and all the while serving as docent and teacher at the local art museum.

Garrick’s son noted changes when his father reached ninety-five. “My father has declined physically and cognitively to a far greater degree than he acknowledges in the questionnaire: gradually over the past 8–10 years, but most precipitously since retiring in 2006 from his acting career. He had worked less for a few years due to his arthritis pain and difficulty remembering lines. When he finally stopped, he has sat in his chair, read books, watched TV, listened to music, ordered more books and music and exercised hardly at all. His work at the theatre had provided a regular dose of social contact and the necessity of pushing himself to analyze the play and character in order to deliver a performance. Both the social interaction and the need to push himself [has now] dwindled to a trickle.”

It’s hard to say with certainty that the decline was really the result of a lack of social and physical activity. Surely the reverse is also possible: that it was more like the collapse of the one-hoss shay after its very long life, and that Garrick, as he approached his own hundredth birthday, gave up his earlier social and physical pleasures as a result. Reality probably lies somewhere in between. Until his death Garrick took only two medications: Aleve for arthritis and Ambien for sleep. At ninety his TICS score was showing some borderline decline, and at age ninety-one, he had a brief trial on Aricept (a drug designed to slow memory loss). When he was ninety-four, his wife, who had become very frail, had to move to assisted living. For a while Garrick remained full of beans, but at ninety-five he began to display some trouble orienting himself. That’s when his son began to worry and came to live with him, doing his shopping, figuring out his taxes, and paying his bills. Garrick himself recognized that he had slowed down—he listed his activities on his last questionnaire as: “Read, watch TV and nap.” But he didn’t seem to be unduly troubled by the change. Daniel Garrick died at ninety-six.

Remember, Garrick was only the first of the men to reach ninety-five, and a gang of mentally alert Study-mates are waiting in the wings to surpass him. Remember, too, that for all of us the play has to end sometime. From sixty to ninety, Garrick had a long and successful run.

Can we learn anything from Daniel Garrick’s life about the keys to graceful aging? Or about what might account for his extraordinary longevity? He certainly didn’t follow most of the conventional “rules” for long life. His parents were not long-lived. (In fact our data indicate that parents’ longevity is only modestly reflected in their children.) He didn’t really start to exercise until he was almost sixty. He smoked a pack of cigarettes a day for twenty years, and nine pipes a day for decades after that. During his sixties, he drank enough that he and his future wife worried about it. Conventional wisdom tells us that we keep young by doing good for others, but he engaged in no community service activities except to become a docent in an art museum—at eighty-six. So when Garrick suddenly began riding his bicycle up to 100 miles a day, was that the cause of his great energy, or was it the effect of some innate natural vigor? Remember, way back in college Garrick had bicycled 5 miles back and forth to class after being up all night, and had gotten top marks for being “well-integrated” and “self-starting,” both traits associated with longevity in the Study.

From the very beginning Garrick had a quality of indomitability, and maybe that was really what enabled him to live so long. Nobody in the Grant Study waited longer or worked harder than he did to get a college education. Most students who lacked financial support received scholarships, but not Garrick. Most of the men who had explicit career ambitions were able to realize them long before they turned fifty. But even during the period when Garrick had resigned himself to being a teacher, he used his spare time to act. He never gave up, and he never gave up hope.

PATHS TO HEALTHY AGING

Sometimes it looks as though cancer and heart attacks are visitations from malicious gods, and that old age is ruled by cruel fate—or at least by cruel genes. So out of control does the whole process of aging often feel that I was relieved when our wealth of prospectively gathered data revealed that some aspects of successful aging—or lack of it—are in fact negotiable.

Twenty years ago Paul Baltes, a leading gerontologist, acknowledged that research has not yet reached a stage where there is good causal (as opposed to correlational) evidence for predicting healthy aging. 18 True, several distinguished ten-to twenty-year prospective studies of physical aging have contributed valuable understanding about the course of old age. 19 But few of the investigators in these studies knew anything about their members before they turned fifty, and only Warner Schaie has followed his subjects for more than twenty-five years. 20 The early facilitating factors in longevity remain unclear. But that doesn't stop people from having strong convictions about them.

For example, we are constantly urged to follow diet, exercise, or neutraceutical regimens that supposedly guarantee us sustained good health. But the formulas are rarely subjected to long-term verification, and the rules keep changing. The cod liver oil of the forties was replaced by the antioxidants of the seventies, which gave way to fish oil again ten years ago. For a while eggs were out, red meat was on everybody's no-no list, and fruit was on the side of the angels. But one of the few thorough studies of nonagenarians has found that 80 percent of them ate red meat regularly all their lives, and only 50 percent ate fruit weekly. 21 Perhaps fruit-eating vegetarians do not survive to ninety. Or perhaps, I, like the diet advocates, am merely revealing my own personal prejudices. Either way, survival is not as simple as the wellness gurus would have us believe. The risk factors that cure-all diets are supposed to counter aren't discrete entities; they are reflections of complex interactions.

My own confrontation with the arbitrariness of longevity prescriptions happened twelve years ago at the Boston Museum of Science. If visitors would punch in all their bad habits, a computer there promised, it would tell them how much longer they had to live. I told the computer that I consumed a half-pound of butter daily and three packages of Camels. I only rarely let a green vegetable touch my lips, I emptied a fifth of Jack Daniels every evening, and I got off the sofa only to change the batteries in the remote. The computer thought for a while, calculated the years that each risk factor was supposedly subtracting from my life, and then informed me that I was dead already, and had been since two years before I was born!

Now I'll grant that this is not a nice way for a person to behave who has given much of his life to collecting, fostering, and analyzing an invaluable hoard of data based on interviews and questionnaires. If the men of the Grant Study had been one-tenth as mendacious as I was. . . . Still, simply summing multiple risks together is very different from watching their complex interactions over real lifetimes, and here's a serious illustration of that.

Harvard sponsored a symposium on aging for its own 350th anniversary. Four experts were asked to describe their research on risk factors. One spoke on exercise, another on obesity, a third on smoking. I spoke on alcoholism. None of us addressed the importance of any factor other than our own, let alone the mischief that the four of them could get into together. Like the computer, we treated the different factors as if each were the only coffin nail in town.

In this chapter, I've tried not to do that. I'm going to recount the history of four separate investigations in which we examined the antecedents of successful aging and longevity. Each led me to a different set of conclusions, and thereby hangs a tale.

FOUR STUDIES OF AGING

Study I, 1978: Mental health and physical health. In 1978, when I was in my early forties and the men were turning fifty-five, I viewed them as over the hill and therefore ripe for a study of successful old age. That's not much of an exaggeration. I thought that they were nearing their apogee, and that I'd better learn what I could before the long decline began. For my first study—the causes of health decline between age forty and age fifty-five—I began with the 189 active members of the Study who had been entirely healthy at forty. Eighty-eight of them had remained that way until fifty-five. What were the differences, I wondered, between them and the 101 men who by that age had either acquired minor chronic illnesses (66) or suffered serious chronic illness or died (35)? In 1979 I believed that Table 7.2 provided the answer. 22

Remember, I was a psychiatrist, not an internist and not an epidemiologist. I wanted an answer to a straightforward question: What was the influence of mental health on physical health? I tested out a number of previously assessed psychological variables: bleak childhood, recourse to mood-altering drugs, major depression, number of psychiatric visits, poor relational skills at midlife, and especially a lack of good adult adjustment between the ages of thirty and forty-seven (as defined by the capacity to work, love, and play, and require no treatment for mental illness). As Table 7.2 shows, all six of these psychological variables were significantly associated with subsequent poor physical health at fifty-five. By then, of the forty-nine men with the bleakest childhoods, seventeen (35 percent) were dead or chronically ill, contrasted with only five (11 percent) of the forty-four men from the warmest childhoods—a very significant difference. On the other hand, by age fifty-five, six (9 percent) of the sixty-six non-smokers and eight (17 percent) of the forty-eight heavy smokers had become chronically ill—a difference of no statistical significance at all.

Table 7.2 Factors Associated with Irreversible Physical Health Decline Before Age 55

Very Significant = $p < .001$; Significant = $p < .01$; NS = Not Significant.

I had made my case; mental illness was very important in hastening the aging process. Every five years I'd trot out these six variables and run them again, looking to see if they were still statistically important factors in looking at physical health. For fifteen years, they were. So far, so good. But alas, like the Science Museum computer, I had not fully controlled for other important risk factors. It took several decades before I statistically separated decline in health after age fifty-five from decline in health before age fifty-five. In retrospect, that should have been a no-brainer. Yet another of the many advantages of longitudinal studies is that they uncover sloppy thinking.

Study II, 2000: Decline in health between fifty-five and eighty. As the men began passing eighty, I was starting to wonder what factors might be associated with the development of dementia. The fact that there is frequently a vascular component to this dread condition led me to establish a five-point scale of vascular risk factors: smoking, alcohol abuse, hypertension, obesity, and Type II diabetes. For the first time I examined the five factors not only individually but as a group; summing risk factors increases their predictive power. When these factors proved to be strongly associated with dementia, I turned them loose on the larger question of physical health in general. And that led to some big surprises.

This time I followed until they were eighty the 177 men who had still been healthy or had had only minor illnesses at fifty-five. And I found with astonishment that the relationship between mental and physical health that had seemed so clear when the men were fifty-five no longer held. As Table 7.3 illustrates, the deterioration of the men's health between ages fifty-five and eighty was very significantly associated with the vascular risk factors, and not at all with the mental health risk factors. Of the 189 healthy men selected at age forty, 103 had no risk factors, and 86 had one or more. Sixty-five (that is, 76 percent) of the latter group were chronically ill or dead by their eightieth year. Of the men with no risk factors, only 45 (44 percent) were dead or disabled. This was a very significant difference.

But the early mental health risk factors predicted nothing about health at eighty. It appears that they did their damage early. The effects of past damage endured, and remained statistically significant even when the effects of alcohol, tobacco use, obesity, and ancestral longevity were controlled. The river of time flowed on, but poor mental health did no further damage. The vascular risk factors, as I documented subsequently (see Table 7.3), were important before fifty-five as well as after, a fact at first unsuspected by this young research psychiatrist. Maturity improves the discernment of researchers, too!

Table 7.3 Factors Associated with Irreversible Physical Health Decline at Different Age Periods

Very Significant = $p < .001$; Significant = $p < .01$; NS = Not S

Study III, 2011: Getting to ninety. The men got older, and I continued to try to identify the variables that allowed some of them to age successfully to ninety. I kept my eye on the vascular risk factors, and again I saw a pair of once-important variables lose their statistical significance as time went by. When the men were between eighty and ninety, the vascular factors as a summed total were still significant. Because so few heavy smokers and alcoholics survived into the ninth decade, however, the significance of those two individual factors faded after eighty. Of the forty-five heavy Study smokers, only sixteen were still alive at eighty and only four at ninety. Only three of the alcohol-dependent men lived until eighty and none until ninety. But the other vascular risk factors told a different story. They continued to affect the men's ability to remain cognitively sharp after eighty; and after eighty dementia was a major

source of health decline. Of the men with no vascular risk factors, 50 percent were still alive at ninety, instead of the expected 30 percent. Of the seventeen men with three or four risk factors, all but one was dead ninety years after his birth.

Table 7.3 sums up the findings of all three of these studies; the first column refers not to the original analysis of the age fifty-five data, but to my later reappraisal, which coded individually for vascular and genetic risks. N in each column refers to living men; one of the reasons that predictive associations change is that certain risk factors, like badly adapted parasites, kill off their hosts, leaving themselves with nothing left to live on.

Study IV, 2012: How to live forever. In 2012, as I write this, sixty-eight of the active Grant Study men are still alive; fifty are still cognitively intact. I wanted to learn what I could about what makes for good health and good mentation over so many years. Ideally this study will be done again ten years from now, by which time we will know exactly when each of the men died. But to bring it all together in a provisional way, I made the best estimate I could of how much longer the men would live, based on past experience. I added seven years to a man's current age if he still had only minor health problems, five years if he was chronically ill, and three years if he was disabled. Obviously there's speculation here, but after the age of ninety there's a limit to how wrong the guess can be. Then I plotted all the men's lifespans against many factors that had been significant in other contexts to see what I could learn about their relationship to longevity. Table 7.4 is the result.

Men with no vascular risk factors lived to an average age of eighty-six. Men with three or more lived to an average age of only sixty-eight. This complex of factors subtracted eighteen years from a man's expected life. The difference in longevity between men with very long- and very short-lived ancestors was seven years—eighty-four years as opposed to seventy-seven—a significant difference, but nothing like that predicted by the vascular risk factors. Surprisingly, parental social class made less than a year's difference. The men with the very best and worst adjustments at forty-seven lived to eighty-five and seventy-seven respectively—an eight-year difference in longevity that was very significant, but again far less than that produced by the vascular risk factors.

A curiosity: the sixteen men with the most combat exposure in World War II died significantly younger than the rest of the sample. They were six times as likely to die from unnatural causes (e.g., murder, suicide) and they were twice as likely to die before eighty. This is another of those unanticipated findings that in the coming years may turn out to be a clue to something unexpected and important.

An interesting independent predictor of longevity, which I recognized only after Friedman and Martin published *The Longevity Project* in 2011, was the presence or absence in college of personality characteristics of perseverance and self-motivation (in the early Study terminology, *Well Integrated* and *Self-Driving*).²³ These traits were significantly associated with longevity; the twenty-three men who manifested both in college were likely to live ten years longer than the men with neither trait (see Table 7.4).

Table 7.4 Factors Associated with Longevity (Mean Age at Death 81.5; N = 237)

Very Significant = $p < .001$; Significant = $p < .01$; NS = Not Significant.

HOW NOT TO LIVE FOREVER

Most of the deaths before fifty-five were associated with factors over which the men had no control. We cannot pick our parents, or control the genes that lead to major depression. The vascular factors that were so strongly implicated in deaths after fifty-five, however, are a different story. They are all now widely accepted as causes of death before eighty, and we were able to confirm this. Using multiple regression analysis (unlike the Boston Museum of Science computer!) to investigate each of the five factors in the context of the other four, we found that each one made an independent contribution to health decline and death.²⁴ As Table 7.3 illustrates, there was a very significant association between these factors and irreversible decline in health in the Grant Study men. The risk factors were of equal importance in the Glueck Inner City cohort and the Terman cohort of gifted women.²⁵ But unlike the previously cited mental health factors, they are causes of early death over which we have considerable control. At the end of the day, good self-care before age fifty—stopping smoking, joining AA, watching the weight, and controlling the blood pressure—made all the difference in how healthy the men were at eighty and ninety.

Education is another factor to be reckoned with when it comes to health; it turned out to be critically important to the health of the Inner City sample (and modestly important for the Grant Study men, whose educational baseline was higher). The average College man who never sought post-graduate education had a life expectancy of seventy-nine years, while for the men with post-graduate educations, life expectancy was eighty-three years—a very significant difference. Years of education were powerfully associated with reduction of all of the five factors affecting vascular health. This was especially true of the Inner City men. This issue will be dealt with more fully in Chapter 10 .

OUT OF OUR HANDS?

However, one interesting piece of conflicting evidence suggests that we still have to reckon with our stars. A recent paper by Thomas Perls and his colleagues on a cohort of 800-plus centenarians suggests that genes are just as important to survival past one hundred as lifestyle. 26 In *Aging Well* I could assert that ancestral longevity was not an important predictor of survival from fifty-five to eighty. But to live past eighty, it appears, ancestral longevity again assumes significance. Another reminder that follow-up must continue as long as life does.

Cancer. Cancer is an increasingly important cause of death after seventy, and this was another factor that often seemed to lie in the men's stars—or, if you prefer, in their genes. With the exception of lung cancer, which is closely tied to smoking, cancer in the Grant Study seemed surprisingly independent of both mental health and the vascular risk factors. In larger samples, however, and in studies focused on specific cancers, environment, dietary and sexual habits, and alcohol abuse can also be demonstrated to be important.

Age and cognition. We used the Telephone Interview for Cognitive Status, or TICS, to measure cognitive skills in men without an existing diagnosis of dementia. 27 In the Grant Study, the factors most associated with a high TICS score appeared to be the absence of vascular risk factors, good vision, college IQ and class rank, exercise at sixty, and, surprisingly, a warm relationship with one's mother when young. This is another curiosity. As we've gone along, I've pointed out that a warm childhood relationship with his mother—not maternal education—was significantly related to a man's verbal test scores, to high salary, to class rank at Harvard, and to military rank at the end of World War II. At the men's twenty-fifth reunion, it looked, to my surprise, as though the quality of a man's relationship with his mother had little effect on overall midlife adjustment. However, forty-five years later, to my surprise again, the data suggested that there was a significant positive correlation between the quality of one's maternal relationship and the absence of cognitive decline. At age ninety, 33 percent of the men with poor maternal relationships, and only 13 percent of men with warm relationships, suffered from dementia.

Dementia, like arthritis, is a curse of longevity. Vital aging at ninety is closely dependent upon the preservation of cognitive faculties, and the best way to achieve this is by minimizing the vascular risk factors. Alzheimer's disease, however, is a special case. It is a major source of health decline after eighty, but, unlike the purely vascular dementias, it seems to be surprisingly independent of the factors that I have been enumerating.

Death before eighty can be avoided to some degree by wise lifestyle choices, but so far the Study has provided no clues as to how to prevent the two greatest and most dreaded sources of total disability after eighty—most cancers and Alzheimer's disease.

WHAT WAS NOT IMPORTANT

Ancestral longevity. Lacking lifetime studies of humans, scientists have studied aging in fruit flies. You can breed and study many generations of fruit flies in a year, and their longevity, it appears, depends heavily upon genes. Therefore, one of the first variables the Study looked at was ancestral longevity.

The myth that ancestral longevity is passed on is hard to test well; this is because most people are either so old that they no longer remember the exact ages of their grandparents' deaths, or so young that they don't yet know when their parents will die. It takes at least a two-generation study to learn from the subjects' parents the age that the grandparents died, and then to follow the subjects until the last of their parents have died as well. (In the Grant Study, the last parent died in 2002, sixty-five years and three generations after the Study began. The Study men themselves were very unreliable informants about the age of death of their grandparents.)

For the College men, we calculated ancestral longevity by averaging the ages of death of the oldest of the first-degree ancestors (parents and grandparents) on the maternal and paternal sides. As I've noted, the longevity of the forty-four men with the longest-lived first-degree ancestors was seven years longer than the longevity of the thirty-six men with the shortest-lived ancestors. That difference was significant, but just barely. Moreover, to my surprise, the average lifespans of the ancestors of the men with the best and worst mental and physical health at eighty were identical. 28 In the Inner City sample as well, the longevity of the men's parents seemed irrelevant to the quality of their own aging at seventy. 29 Fruit flies are clearly not always good models for human aging. Obviously, specific genes are very important in predicting the specific illnesses that shorten life, and soon the precise genes that facilitate longevity may be discovered. In the meantime, however, the McArthur twin studies and investigators using the Swedish Twin Registry confirm our finding that variance in longevity cannot simply be attributed to genetic inheritance. 30

Psychosomatic stress. When the Study began, psychosomatic medicine was in high fashion. Hans Selye had shown the world that stress could kill, and psychoanalytic theories about the role of emotions in medical illness were all the rage, attributing peptic ulcers to repressed anger or longing for love. 31 It would be decades before the medical profession brought itself to accept that the most common cause of duodenal ulcers is a gram-negative bacterium of the genus *Helicobacter*.

Besides, many people do feel physically sick under stress. They have headaches; they can't sleep; their stomachs ache; they get itchy; they're running to the bathroom all the time. This observation suggests an attractive hypothesis, to which in the 1960s I earnestly subscribed: that individuals who experience stress psychosomatically in midlife would suffer poor physical health in old age. 32 When I joined the Grant Study in 1966, I was thrilled to be in possession of prospective Study data that I suspected would prove this hypothesis.

A secondary hypothesis in psychosomatics, to which I also subscribed, held that individuals have characteristic "target organs" in which they experience stress. 33 This proposition was based on the observation that one person's place of experiencing somatic symptoms under stress may be quite different from another's, and it was much emphasized in my residency training. Theoreticians anticipated that the organ affected by stress would be the same one in which signs of psychosomatic illness appear.

A third related hypothesis, one that I felt certain of proving, was the implicit assumption that the development of "psychosomatic" illnesses (for example, colitis, asthma, and hypertension) reflect more psychopathology than the development of "real" illnesses (such as diabetes, myocardial infarction, and osteoarthritis). Indeed, many of the mental health screening tests of the 1950s and '60s, some of which, like the Minnesota Multiphasic Personality Inventory, are still in use today, use the presence of multiple physiological symptoms under stress as indicators of emotional illness.

In the 1970s I finally analyzed the data to test these three psychosomatic hypotheses so favored by me and other armchair speculators. My intention was to demonstrate 1) that psychosomatic illness leads to accelerated aging; 2) that "target organs" for stress (the stomach, the lungs, and so on) are real and remain stable over time; and 3) that psychosomatic illness is a reliable hallmark of mental illness. Longitudinal study proved them all wrong.

(Note that the evidence for this belief system had been derived over the years from retrospective data. Furthermore, the source of the data was patients who presented themselves repeatedly for medical attention—a group likely to have mental health issues. 34 The Grant Study, on the other hand, was a prospective study of men who explicitly were not patients.)

Over the years, the Study men were systematically asked where in their bodies they experienced emotional stress. Decades of follow-up revealed that this site varied considerably over time. 35 The idea of stable target organs (hypothesis II) was not supported.

Nor did the number of physical symptoms under stress before age fifty predict physical health at age seventy-five or at ninety. If you wait a few decades, people often recover from psychosomatic illnesses. And at age eighty, the physical health of the men with multiple apparently psychosomatic illnesses was actually, if insignificantly, better than that of men with none. Hypothesis I was not confirmed either.

Hypothesis III, the supposition that there's a connection between psychosomatic illness and mental health, didn't fare any better than the other two. The Study has carefully followed the men's objective physical health from the age of forty,

when it was still very good, up through the present or until their deaths. By fifty, over half of the Study men had required medical treatment for conditions thought by some physicians to be psychosomatic: hypertension, respiratory allergies, ulcer, colitis, and chronic musculoskeletal complaints. Admittedly, these five disorders do not define psychosomatic illness conclusively, but they make a good start. At age forty-seven, after almost thirty years of observation, the men were ranked for emotional health. There was no correlation with the number of psychosomatic complaints. The number of psychosomatic illnesses that the men suffered didn't predict mental health at age sixty or eighty, either. As a faithful believer in psychosomatic medicine, I was profoundly disappointed.

We also found that the childhoods of the men who experienced relatively few physical symptoms under stress and no psychosomatic illnesses were no warmer in blind ratings than those of men cursed with a great deal of psychosomatic symptomatology. 36 As I've noted, in our first study of aging, premorbid psychopathology did correlate with early onset of chronic (real) physical illness, but only up to the age of fifty-five. We had some middle-aged Study men who frequently visited medical doctors and who took more than five or more sick days a year. They displayed our distinguishing markers for mental health issues—drinking, psychiatric intervention, use of tranquilizers and antidepressants, and so on—and in this they resembled the self-selected populations of the old retrospective studies. In other words, people with mental health issues brought more illnesses of all kinds—real and imagined—to the attention of the medical community.

When in 1970 I first presented these findings at a national psychosomatic meeting, I was met with outrage and disbelief as a traitor to dynamic psychiatry. Today, my findings would be regarded as unremarkable. Over the last forty years, a medical model of psychosomatics has displaced the earlier one, thanks in part to the *Helicobacter* story and the fading influence of psychoanalysis in academic psychiatry. My research life, and the Study's, spanned both eras.

Cholesterol. Magazines would lose valuable advertising revenues, and probably readers, if they talked too much about really significant risks to health—like Virginia Slims and wine coolers. But it's OK to fuss about cholesterol, because the butter and egg lobbies don't advertise in *Cosmopolitan*. Moreover, as the TV ads tell us, your cholesterol can be magically lowered by statins, even if you don't eat less or exercise more. So the war on cholesterol would appear to be a win-win-win situation—for patient, for doctor, and for the pharmaceutical industry.

Countless studies show that the ratio between high- and low-density lipoproteins (HDLs and LDLs) is important and that reducing LDL levels is good for the heart. But in neither the College nor the Inner City sample did average cholesterol levels at age fifty distinguish the men who lived to ninety from those who died before eighty. The estimated age of death of the fifty-eight Grant Study men with cholesterol levels below 206 mg/dl was eighty-three. The estimated age of death of the fifty-seven men with cholesterol levels over 254 mg/dl was eighty-one, which was not a statistically significant difference. This finding has been confirmed by much larger and more representative studies. 37

It's situations like these that persuaded me to include my experiment with the Boston Museum of Science computer—not as entertainment, but as an alert. To understand longevity we need longitudinal images, not snapshots. The more sacred our cows, the more they need longitudinal testing. This is one of the ones we were able to test.

Bleak childhood. Alas, we cannot choose our families. Without asking permission they endow us with their genes. They also bathe us in their warmth and riches—or parch us with emotional and financial poverty. Many aspects of childhood are important to aging, as I've described in Chapter 4. But most of them do not predict length of life. Parental social class, stability of parental marriage, parental death in childhood, and IQ (at least in our samples with their two different but limited ranges) were not important to longevity. The men from the warmest childhoods lived only a year and a half longer than the men from the bleakest childhoods, a difference that was not statistically significant.

Vital affect and general ease in social relationships. These were the two personality traits most highly valued by the original Study investigators, and for the Harvard cohort, they correlated highly with good psychosocial adjustment in college and in early adulthood. 38 But they, too, failed to predict healthy aging.

CONCLUSION

After seventy-five years of collecting, analyzing, and reanalyzing our data on aging, what have we learned from all this? First, we've been reminded again and again that a strong association does not necessarily imply cause. Snow is associated with winter, but snow does not cause winter. Heavy smoking is strongly associated with fatal automobile

accidents, but not because drivers take their eyes off the road to search for the cigarette lighter. The association is the result of a third factor—alcoholism, which very significantly increases the likelihood both of heavy smoking and of fatal accidents. 39 The seventy-year duration of the Grant Study have been invaluable in allowing us to draw lines between causes and associations.

For example, there is a strong association between exercise and physical health, which most of us understand to mean that exercise causes good health. 40 But might it not be the other way around? Healthy people enjoy exercise. Among the College sample, exercise at age sixty correlated more highly with health at age fifty-five than with health at eighty. Health at fifty very significantly predicted exercise at eighty, and health at sixty significantly predicted exercise at eighty. In other words, health predicted exercise at all ages, but exercise did not predict health in later years.

Admittedly, exercise at age thirty did significantly predict health at fifty-five and sixty, and exercise at sixty predicted health at age seventy through eighty-five, although not significantly. Presumably the exercise mavens must not be written off entirely. But it's important to keep in mind—as the Museum of Science computer didn't—that everything affects everything else, and that some things are horses and others are carts. When it comes to physical aging, alcohol abuse and education are looking more and more like horses. And, as Table 7.3 documents to my sorrow, after thirty years of betting the farm that maturity of defenses (the involuntary coping mechanisms that I'll discuss in the next chapter), would be the horses that pull us to late-life physical health, longitudinal study has proved me wrong. He who lives by the sword dies by the sword.

Social supports are often assigned a causal role in successful aging. In his classic review of the evidence, however, sociologist James House acknowledged that almost no attention has been paid to social supports as a dependent variable. 41 That is, social supports may be the result of the very variables that they are supposed to be causing. In the prolonged prospective view of the Grant Study, social supports at age seventy were strongly associated with the pre-age-fifty protective health factors identified in Table 7.2, yet only weakly associated with longevity (Table 7.3). In other words, good health predicts good social support better than good social support predicts future health. Indeed, good social supports in old age may be in large part a result of earlier habits that preserve physical health. 42

There was a powerful association between the absence of cigarette and alcohol abuse before fifty and good social supports at seventy. Specifically, there were some with good supports among the Grant Study men who abused alcohol or cigarettes. There were also men with poor social supports but no tobacco or alcohol abuse. In these asymmetric pairs, the risky habits of alcohol and cigarette use were the horse, and social supports the cart. Study men with good social supports but risky habits suffered health just as poor as men with poor social supports and risky habits. Men with poor social supports but good habits enjoyed good health almost as often as men with good social supports and good habits.

My point here is not that love and exercise aren't good for you. For most of us, the more social supports we have in our old age the happier we'll be, and the more exercise we get the better we'll feel and look. I only wish to underscore that the etiology of successful aging is multifactorial in ways that self-help books and cross-sectional studies do not necessarily take into account. Furthermore, the Grant Study demonstrates, if you follow lives long enough, the risk factors for healthy life adjustment change. There is an age to study the relation between mental and physical illness and an age to ignore it. There is an age to curse your arthritis and an age to appreciate it as the price of having survived to attend your granddaughter's wedding.

Please note too that I report these contrary findings as an investigator who claimed in Chapter 1 that love is the root of all blessings, and who believed, and for many years tried to prove, that mental health caused physical health. I've had to give up on that second claim, alas; my own worldview is as vulnerable to the upheavals of longitudinal follow-up as anyone else's. But I will retreat no farther than that. I am wiser than the Museum of Science computer, and I have grasped clearly that when it comes to healthy aging, everything really is connected to everything else. A happy old age requires both physical health and mental health. For mental health, love is a necessity. So is being alive. So is being able to think straight. We need physical and cognitive competence to build the social surrounds that give us love and support later on, and it is love and support that encourage us to care for ourselves well and keep ourselves healthy, even when the going gets rough. "Button Up Your Overcoat" isn't so far off the mark after all. The ninety-year-olds of the Grant Study took good care of themselves and of their important relationships. And for the most part, they've been very happy to be alive.